

Andrew Yuan

📍 New York ✉ aycooper2022@gmail.com 🌐 My Website 📄 GitHub

Education

The Cooper Union

Expected Graduation 2026

B.Eng: Electrical Engineering - Computer Eng. Track

Minors: Mathematics, Computer Science

- **GPA:** 3.93/4.0 — **Major/Minor GPAs:** 4.0/4.0
- **Relevant Coursework:** **denotes graduate level*
ECE: Bayesian Machine Learning*, Financial Signal Processing*, Digital Signal Processing, Probability Models & Stochastic Processes, Communication Networks, Operating Systems, Data Structures & Algorithms,
Math: Convex Optimization*, Linear Algebra, Real Analysis, Modern Algebra, Differential Equations
- **Languages:** C/C++, Python, MATLAB, L^AT_EX
- **Tools/Libraries:** Git, NumPy, Pandas, GPyTorch, BoTorch, sci-kit image, SciPy, LTspice, Logisim-Evolution

Publications

Kernel Assessment in High Dimensional Bayesian Optimization

(Manuscript In Progress)

Abhigya Koirala, **Andrew Yuan**, Jimmy Risk

Presentations

Cal Poly Pomona Creative Activities & Research Symposium

Summer 2025

Oral Presentation: Kernel Assessment in High Dimensional Bayesian Optimization

Abhigya Koirala, **Andrew Yuan**

Research Experience

Large Language Model-based Psychotherapy Evaluations

New York, NY

Independent Research

November 2025 – Present

- Conducted a literature review on the use of large language models (LLMs) and multimodal LLMs for psychotherapy under Dr. Xiaofan (Fred) Jiang of Columbia University.
- Reviewed clinician-rated evaluation benchmarks for open-ended mental health dialogue and designed prompts to simulate multi-turn psychotherapy interactions, qualitatively assessing LLM conversational behavior and responses.

NSF REU: Kernel Engineering in Gaussian Processes

Pomona, CA

Cal Poly Pomona

June 2025 – August 2025

- Selected as part of the Computational and Applied Mathematics Program funded by the NSF Research Experience for Undergraduates at Cal Poly Pomona under Dr. Jimmy Risk.
- Developed a reproducible experimental framework with automated training loops and batched evaluations using BoTorch and GPyTorch to benchmark Gaussian process kernels for high dimensional Bayesian Optimization (BO)
- Implemented a Python data-recording framework for kernel hyperparameters and BO performances to generate data visualizations for comparison across experiments.
- Analyzed asymptotic behavior of kernel hyperparameters, examining results relative to machine epsilon to identify sources of *gradient vanishing* and model degeneration.
- Authored technical report proposing alternate kernel choices to remedy *gradient vanishing* and improve high dimensional BO performance (*Research manuscript extending technical report in progress*).

Projects

Single Image Super-Resolution with Gaussian Processes

[Link](#) 

- Implemented a Single Image Super Resolution framework using GPyTorch and Sci-kit Image, applying Gaussian Process priors to model and interpolate spatial and RGB value dependencies of pixels.
- Adapted algorithm to handle independent RGB channel modeling, designing GP model training setup including patch extraction and reconstruction steps for color images.
- Compared image reconstruction quality across different kernels with baseline interpolation methods using PSNR and SSIM metrics.

Optimization of Subclass Damage Profiles in AdventureQuest Game

[Link](#) 

- Formulated an integer linear programming (ILP) and integer quadratic programming (IQP) models to evaluate

optimal resource allocations under linear constraints.

- Implemented and solved ILP/IQP models in Python using Pyomo, GLPK, and Gurobi solvers, incorporating ability multipliers, hybrid classes, and multi-turn effects.
- Analyzed linear-programming relaxations and corresponding dual problems to compare between discrete and continuous resource allocation strategies.

Analog Audio Color Organ Device

[Link](#) 

- Designed a full analog audio-processing pipeline with input volume control, signal-amplification, multi-band filtering, and LED-driver output stages for line-in signals.
- Implemented band-pass filtering and detection stages using op-amps and envelope detectors, conditioning each band's amplitude to drive LEDs in real time to visualize audio spectral content.

Hall-Effect Analog Input Multi-Purpose Interface

In Progress

- Senior Capstone Project prototyping a Hall-effect analog-input keypad to provide precise and versatile input interface adaptable to consumer applications.
- Developing embedded firmware and digital filtering on the Raspberry Pi Pico 2 (RP2350), alongside a host-side software application to enable user-configurable actuation thresholds

Teaching and Mentorship

Math Department Tutor and Grader

Fall 2024-Present

The Cooper Union

- Individual one-on-one and weekly drop-in tutoring sessions for 5+ students in: Calculus I-II, Vector Calculus, Linear Algebra, Probability, and Differential Equations.
- Evaluated and provide feedback for 60+ students in Calculus I-II and Intro to Linear Algebra courses.
- Met weekly with the course instructor to review student performance and offer feedback on common difficulties and instructional adjustments.

IEEE Cooper Union Mentor

Fall 2024-Present

The Cooper Union

- Mentored first-year Electrical Engineering and Computer Science students, providing tutoring for Electrical and Computer Engineering courses and advising on course planning and professional development.

Extracurriculars

International Finance Student Association Chapter Co-Founder

Fall 2024-Present

The Cooper Union

- Co-founded the first North American chapter of International Finance Student Association at Cooper Union, growing membership to 30+ active participants, surpassing the average club size of 10.
- Organized guest-speaker events and workshops with a budget of \$1000 to promote professional development and provide networking opportunities.

Professional Badminton Athlete

2012-Present

USA Badminton Association

- **USA Adult National Team** - FISU World University Games (2025)
- **USA Jr. National Team** - World Junior Championships (2022), Pan American Junior Championships (2014-2022)
- Former Junior National Rank #1, Junior National Champion, Pan American Junior Champion

Awards and Honors

- Howard Flagg EE'75 Memorial Prize Scholarship (Fall 2025) - EE'26 Class Rank 1
- Jesse Sherman EE'33 Book Award Scholarship (Fall 2024, Spring 2025)
- Tau Beta Pi Honor Society
- Eta Kappa Nu Honor Society
- Dean's List 2022-Present (All semesters)
- CU Innovator's Merit Scholarship 2022-2026